

Se Min Kong

SOFTWARE | VISION | ROBOTICS

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PROFILE

Software engineering graduate who enjoys learning new technologies and turning ideas into working systems. Project experience spans computer vision, ROS 2, backend services, and hardware integration, with a focus on clear interfaces, observable behavior, and end-to-end implementation.

EDUCATION

Soongsil University | College of IT, School of Software Mar 2020 - Feb 2026
Bachelor of Engineering in Software; Major in Artificial Intelligence & Big Data; Convergence Major in Big Data
GPA 3.61/4.50 | Graduated 20 Feb 2026
Dulwich College Suzhou (DCSZ) | Primary & Middle School, Suzhou, China 2008 - 2012

TECHNICAL SKILLS

Languages & Core: Python, C++, TypeScript
AI & Vision: PyTorch, Ultralytics YOLO, MediaPipe Hands, OpenCV, Transformers, CUDA
Robotics & Systems: ROS 2, DYNAMIXEL XL330, Jetson, Raspberry Pi 5, Intel RealSense, Dobot, Isaac Sim / Isaac Lab
Backend & Tools: FastAPI, Django, WebSocket, React, Node.js, Socket.io, Git, Docker, Linux (Ubuntu)

EXPERIENCE

Samsung SW-AI Academy For Youth (SSAFY) | Trainee, Robotics Track Jan 2026 - Present

- Completing an intensive SW-AI curriculum centered on robotics, computer vision, ROS 2, and integrated hardware-software projects.

Republic of Korea Army | Tactical Communications Equipment Operator/Maintainer May 2021 - Nov 2022

- Completed mandatory military service and was discharged as a Sergeant.

SELECTED PROJECTS

THING Human-Mimetic Robot Hand | Team Project, Robotics & Edge AI Jul 2026 - Aug 2026
ROS 2 | MediaPipe Hands | OpenCV | Jetson | Raspberry Pi 5 | DYNAMIXEL XL330 | React | [Case Study](#)

- Team-built a human-mimetic tendon hand that maps 21 MediaPipe landmarks to a seven-axis HandCommand and drives seven DYNAMIXEL XL330 actuators through ROS 2.
- Integrated Jetson perception, Raspberry Pi 5 command mediation, Command Guard and GPIO E-Stop, React monitoring, and rosbag2-to-EC2 experiment records.

AQIS for Smart Factory | Team Lead, Full-Stack & Robot Integration May 2026 - Jun 2026
ROS 2 | FastAPI | WebSocket | React | RealSense | Dobot | YOLOv5 | [Repository](#)

- Led full-stack and robot integration on a two-person team for a smart-factory system: RealSense + YOLOv5 inspection triggered conveyor stops and Dobot vacuum pick-and-place.
- Built the RealOps dashboard, FastAPI REST/WebSocket services, ROS 2 bridge, device adapters, and LLM command integration for live monitoring and control.

Brain Tumor MRI Classification & Segmentation | Personal Project Feb 2026 - Apr 2026
Python | PyTorch | Ultralytics YOLO11 | OpenCV | NumPy | [Repository](#)

- Built a unified YOLO11 classification and instance-segmentation pipeline for four MRI classes, combining diagnosis and tumor-mask visualization in one output.
- Automated grayscale mask-to-polygon label conversion; reported 99.4% Top-1 accuracy and 92.7% Mask mAP50 in project evaluations.

Alkkagi.io | Personal Project Mar 2026 - Apr 2026
React 19 | TypeScript | Node.js | Express | Socket.io | [Repository](#)

- Built a real-time multiplayer browser game with a React/TypeScript frontend and Node.js/Express/Socket.io backend.
- Implemented a server-authoritative 60 FPS physics engine for elastic collisions, friction, momentum, position correction, and dynamic mass/radius scaling.

AWARDS & CERTIFICATIONS

2025 IT Project Pro League - Encouragement Award | Soongsil University Spartan SW Education Center 22 Nov 2025
Software Competition - Gold Prize | Soongsil University SW
Capstone Design Competition - Encouragement Award | Soongsil University SW
OPIc English - IH (Intermediate High) | Tested 5 Oct 2025 | Valid through 4 Oct 2027